

Homework 3

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1 Problem 1

The FTCS discretization of the diffusion equation is

$$T_j^{n+1} = T_j^n + \frac{\kappa \Delta t}{(\Delta x)^2} (T_{j+1}^n - 2T_j^n + T_{j-1}^n).$$

It is useful to define

$$\alpha \equiv \frac{\kappa \Delta t}{(\Delta x)^2},$$

so that

$$T_j^{n+1} = \alpha T_{j-1}^n + (1 - 2\alpha)T_j^n + \alpha T_{j+1}^n.$$

(a)

Writing the update as

$$\mathbf{T}^{n+1} = A\mathbf{T}^n,$$

one row of the matrix A , away from the boundaries, is

$$A_{j,:} = \begin{pmatrix} \cdots & \alpha & 1 - 2\alpha & \alpha & \cdots \end{pmatrix}.$$

Thus, the three nonzero elements in an interior row are

$$A_{j,j-1} = \alpha, \quad A_{j,j} = 1 - 2\alpha, \quad A_{j,j+1} = \alpha.$$

(b)

For the von Neumann stability analysis, assume a Fourier mode of the form

$$T_j^n = \xi^n(k) e^{ikj\Delta x}.$$

Substituting this into the FTCS update gives

$$\xi^{n+1} e^{ikj\Delta x} = \xi^n e^{ikj\Delta x} + \alpha \left[\xi^n e^{ik(j+1)\Delta x} - 2\xi^n e^{ikj\Delta x} + \xi^n e^{ik(j-1)\Delta x} \right].$$

Dividing by $\xi^n e^{ikj\Delta x}$ gives

$$\xi = 1 + \alpha \left(e^{ik\Delta x} - 2 + e^{-ik\Delta x} \right).$$

Using

$$e^{i\theta} + e^{-i\theta} = 2 \cos \theta,$$

this becomes

$$\xi = 1 + 2\alpha (\cos(k\Delta x) - 1).$$

The identity

$$\cos \theta - 1 = -2 \sin^2 \left(\frac{\theta}{2} \right)$$

then gives

$$\boxed{\xi(k) = 1 - 4 \frac{\kappa \Delta t}{(\Delta x)^2} \sin^2 \left(\frac{k \Delta x}{2} \right)}.$$

(c)

For the method to be stable, every Fourier mode must satisfy

$$|\xi(k)| \leq 1.$$

Since $\alpha > 0$, the largest possible value of ξ is already 1. The lower bound therefore gives the timestep restriction,

$$1 - 4\alpha \sin^2\left(\frac{k\Delta x}{2}\right) \geq -1.$$

The most restrictive mode occurs when

$$\sin^2\left(\frac{k\Delta x}{2}\right) = 1.$$

Therefore,

$$1 - 4\alpha \geq -1,$$

so that

$$4\alpha \leq 2,$$

or

$$\alpha \leq \frac{1}{2}.$$

Substituting the definition of α ,

$$\frac{\kappa\Delta t}{(\Delta x)^2} \leq \frac{1}{2},$$

which gives

$$\boxed{\Delta t \leq \frac{(\Delta x)^2}{2\kappa}}.$$

(d)

For a fixed physical domain,

$$\Delta x = \frac{x_{\max} - x_{\min}}{J}.$$

Doubling the spatial resolution means replacing J by $2J$, so

$$\Delta x \rightarrow \frac{\Delta x}{2}.$$

Since the stable timestep scales as

$$\Delta t_{\max} \propto (\Delta x)^2,$$

the new timestep must satisfy

$$\Delta t_{\max} \rightarrow \frac{1}{4} \Delta t_{\max}.$$

Therefore, doubling the spatial resolution requires reducing the timestep by a factor of

$$\boxed{4}.$$

2 Problem 2

The fully implicit scheme evaluates the spatial derivative using the new temperature values at time $n + 1$,

$$T_j^{n+1} = T_j^n + \alpha \left(T_{j+1}^{n+1} - 2T_j^{n+1} + T_{j-1}^{n+1} \right),$$

where again

$$\alpha = \frac{\kappa \Delta t}{(\Delta x)^2}.$$

(a)

Collecting all of the $n + 1$ terms on the left gives

$$-\alpha T_{j-1}^{n+1} + (1 + 2\alpha) T_j^{n+1} - \alpha T_{j+1}^{n+1} = T_j^n.$$

For boundary conditions

$$T_0 = T_J = 0,$$

only the $J - 1$ interior temperatures need to be solved for. Defining

$$\mathbf{T}_{\text{int}}^n = \begin{pmatrix} T_1^n \\ T_2^n \\ \vdots \\ T_{J-2}^n \\ T_{J-1}^n \end{pmatrix},$$

the update can be written as

$$B\mathbf{T}_{\text{int}}^{n+1} = \mathbf{T}_{\text{int}}^n,$$

with the tridiagonal matrix

$$B = \begin{pmatrix} 1+2\alpha & -\alpha & 0 & \cdots & 0 \\ -\alpha & 1+2\alpha & -\alpha & \ddots & \vdots \\ 0 & -\alpha & 1+2\alpha & \ddots & 0 \\ \vdots & \ddots & \ddots & \ddots & -\alpha \\ 0 & \cdots & 0 & -\alpha & 1+2\alpha \end{pmatrix}.$$

The boundary rows do not need to be included because the two wall temperatures remain fixed at zero.

(b)

Using the Fourier mode

$$T_j^n = \xi^n e^{ikj\Delta x},$$

the implicit update becomes

$$\xi^{n+1} e^{ikj\Delta x} = \xi^n e^{ikj\Delta x} + \alpha \left[\xi^{n+1} e^{ik(j+1)\Delta x} - 2\xi^{n+1} e^{ikj\Delta x} + \xi^{n+1} e^{ik(j-1)\Delta x} \right].$$

Dividing by $\xi^n e^{ikj\Delta x}$ gives

$$\xi = 1 + \alpha\xi \left(e^{ik\Delta x} - 2 + e^{-ik\Delta x} \right).$$

As in Problem 1,

$$e^{ik\Delta x} - 2 + e^{-ik\Delta x} = -4 \sin^2 \left(\frac{k\Delta x}{2} \right),$$

so

$$\xi = 1 - 4\alpha\xi \sin^2 \left(\frac{k\Delta x}{2} \right).$$

Collecting the factors of ξ ,

$$\xi \left[1 + 4\alpha \sin^2 \left(\frac{k\Delta x}{2} \right) \right] = 1.$$

Therefore,

$$\xi(k) = \frac{1}{1 + 4\frac{\kappa\Delta t}{(\Delta x)^2} \sin^2 \left(\frac{k\Delta x}{2} \right)}.$$

(c)

For

$$\kappa > 0, \quad \Delta t > 0,$$

the denominator of ξ is always greater than or equal to one. Therefore,

$$0 < \xi(k) \leq 1$$

for every wavenumber and every positive timestep.

Thus the fully implicit method is unconditionally stable and there is no finite timestep restriction from stability,

$$\Delta t \text{ is unrestricted by stability.}$$

Very large timesteps may still give a poor approximation to the time evolution, but they do not make the numerical solution unstable.

(d)

Taking the limit

$$\Delta t \rightarrow \infty$$

also gives

$$\alpha \rightarrow \infty.$$

For every nonconstant Fourier mode,

$$\xi(k) = \frac{1}{1 + 4\alpha \sin^2(k\Delta x/2)} \rightarrow 0.$$

Since the temperatures at both boundaries are fixed to zero, the equilibrium solution of the diffusion equation is

$$T(x) = 0.$$

Therefore, in the large-timestep limit,

$$\boxed{T_j \rightarrow 0}$$

at every interior grid point.

3 Problem 3

The Crank–Nicholson method averages the explicit and implicit spatial derivatives. Its update rule is therefore

$$\frac{T_j^{n+1} - T_j^n}{\Delta t} = \frac{\kappa}{2(\Delta x)^2} \left[T_{j+1}^n - 2T_j^n + T_{j-1}^n + T_{j+1}^{n+1} - 2T_j^{n+1} + T_{j-1}^{n+1} \right].$$

Using

$$\alpha = \frac{\kappa \Delta t}{(\Delta x)^2},$$

this can also be written as

$$T_j^{n+1} = T_j^n + \frac{\alpha}{2} \left[T_{j+1}^n - 2T_j^n + T_{j-1}^n + T_{j+1}^{n+1} - 2T_j^{n+1} + T_{j-1}^{n+1} \right].$$

Collecting the new-time terms on the left gives

$$-\frac{\alpha}{2} T_{j-1}^{n+1} + (1 + \alpha) T_j^{n+1} - \frac{\alpha}{2} T_{j+1}^{n+1} = \frac{\alpha}{2} T_{j-1}^n + (1 - \alpha) T_j^n + \frac{\alpha}{2} T_{j+1}^n.$$

(a)

Again assume

$$T_j^n = \xi^n e^{ikj\Delta x}.$$

Substitution into the Crank–Nicholson equation gives

$$\xi - 1 = \frac{\alpha}{2} \left[e^{ik\Delta x} - 2 + e^{-ik\Delta x} + \xi \left(e^{ik\Delta x} - 2 + e^{-ik\Delta x} \right) \right].$$

Factoring gives

$$\xi - 1 = \frac{\alpha}{2} (1 + \xi) \left(e^{ik\Delta x} - 2 + e^{-ik\Delta x} \right).$$

Using

$$e^{ik\Delta x} - 2 + e^{-ik\Delta x} = -4 \sin^2 \left(\frac{k\Delta x}{2} \right),$$

we obtain

$$\xi - 1 = -2\alpha(1 + \xi) \sin^2 \left(\frac{k\Delta x}{2} \right).$$

Let

$$s^2 = \sin^2 \left(\frac{k\Delta x}{2} \right).$$

Then

$$\xi - 1 = -2\alpha s^2 - 2\alpha s^2 \xi,$$

so

$$\xi(1 + 2\alpha s^2) = 1 - 2\alpha s^2.$$

Therefore,

$$\boxed{\xi(k) = \frac{1 - 2\frac{\kappa\Delta t}{(\Delta x)^2} \sin^2 \left(\frac{k\Delta x}{2} \right)}{1 + 2\frac{\kappa\Delta t}{(\Delta x)^2} \sin^2 \left(\frac{k\Delta x}{2} \right)}}.$$

(b)

Define

$$q = 2\alpha \sin^2 \left(\frac{k\Delta x}{2} \right),$$

where $q \geq 0$. The amplification factor becomes

$$\xi = \frac{1 - q}{1 + q}.$$

Since

$$|1 - q| \leq 1 + q$$

for every $q \geq 0$,

$$|\xi| \leq 1$$

for every wavenumber and every positive timestep.

Thus, like the fully implicit method, the Crank–Nicholson method is unconditionally stable,

Δt is unrestricted by stability.

For sufficiently large timesteps, ξ can become negative for the highest-frequency modes, so those modes may alternate in sign between timesteps. Their magnitude still remains less than or equal to one, however, so this does not violate the stability condition.

4 Problem 4

I implemented the FTCS, fully implicit, and Crank–Nicholson methods from Problems 1–3 and compared them to the analytic solution for a uniform rod initially at

$$T(x, 0) = 1$$

with its ends held at

$$T(x_{\min}, t) = T(x_{\max}, t) = 0.$$

For a domain of length

$$L = x_{\max} - x_{\min},$$

the analytic solution can be written as the Fourier series

$$T(x, t) = \sum_{\substack{m=1 \\ m \text{ odd}}}^{\infty} \frac{4}{m\pi} \sin\left(\frac{m\pi(x - x_{\min})}{L}\right) \exp\left[-\kappa\left(\frac{m\pi}{L}\right)^2 t\right].$$

For the numerical comparison, I evaluated this series using the first 100 odd modes. The tests use the default conductivity

$$\kappa = 1.$$

For all three numerical methods I used

$$\alpha = \frac{\kappa \Delta t}{(\Delta x)^2}.$$

The FTCS method directly applies

$$T_j^{n+1} = T_j^n + \alpha (T_{j+1}^n - 2T_j^n + T_{j-1}^n).$$

For the fully implicit method, the new temperature values were obtained by solving the tridiagonal system

$$-\alpha T_{j-1}^{n+1} + (1 + 2\alpha)T_j^{n+1} - \alpha T_{j+1}^{n+1} = T_j^n.$$

For Crank–Nicholson, I solved

$$-\frac{\alpha}{2}T_{j-1}^{n+1} + (1 + \alpha)T_j^{n+1} - \frac{\alpha}{2}T_{j+1}^{n+1} = \frac{\alpha}{2}T_{j-1}^n + (1 - \alpha)T_j^n + \frac{\alpha}{2}T_{j+1}^n.$$

The implicit systems with fixed boundaries were solved using a banded tridiagonal solver. The implementations also support the periodic boundary condition used by the other provided initial conditions.

$$J = 20$$

For

$$J = 20,$$

the spatial resolution is

$$\Delta x = \frac{1}{20} = 0.05.$$

From the FTCS stability condition derived in Problem 1,

$$\Delta t \leq \frac{(\Delta x)^2}{2\kappa},$$

the largest stable timestep is

$$\Delta t = \frac{0.05^2}{2} = 1.25 \times 10^{-3}.$$

This gives

$$\alpha = 0.5$$

and requires

800 timesteps

to integrate from $t = 0$ to $t = 1$.

All three methods remained stable. The mean RMS differences from the analytic solution over the full run were

$$\begin{aligned} \text{FTCS} &: 1.03 \times 10^{-3}, \\ \text{Implicit} &: 8.46 \times 10^{-4}, \\ \text{Crank-Nicholson} &: 1.57 \times 10^{-4}. \end{aligned}$$

Crank-Nicholson gave the smallest error of the three methods for this test.

$J = 40$ with the same timestep

I next doubled the spatial resolution to

$$J = 40,$$

while keeping

$$\Delta t = 1.25 \times 10^{-3}.$$

The new grid spacing is

$$\Delta x = \frac{1}{40} = 0.025,$$

so that

$$\alpha = \frac{1.25 \times 10^{-3}}{0.025^2} = 2.$$

This is well above the FTCS stability limit

$$\alpha \leq \frac{1}{2}.$$

As expected, the FTCS solution became unstable and grew until numerical overflow occurred. The fully implicit and Crank–Nicholson solutions remained stable because neither method has a stability-imposed upper limit on Δt .

Their mean RMS differences were

$$\begin{aligned} \text{Implicit} &: 8.21 \times 10^{-4}, \\ \text{Crank–Nicholson} &: 1.05 \times 10^{-4}. \end{aligned}$$

$J = 40$ with a stable FTCS timestep

For $J = 40$, the largest stable FTCS timestep is

$$\Delta t = \frac{0.025^2}{2} = 3.125 \times 10^{-4}.$$

This is a factor of four smaller than the stable timestep for $J = 20$, as expected from the scaling

$$\Delta t_{\max} \propto (\Delta x)^2.$$

The integration now requires

3200 timesteps

to reach $t = 1$, with

$$\alpha = 0.5.$$

With this timestep, all three methods were stable. The mean RMS differences were

$$\begin{aligned} \text{FTCS} &: 2.80 \times 10^{-4}, \\ \text{Implicit} &: 2.32 \times 10^{-4}, \\ \text{Crank–Nicholson} &: 4.31 \times 10^{-5}. \end{aligned}$$

Increasing the spatial resolution therefore required a fourfold reduction in the FTCS timestep, while the implicit methods remained stable without this restriction. Crank–Nicholson produced the smallest error in each stable comparison.

See `problem4.ipynb`.